

Term Paper

Computing the Impact of Investment in Renewable Energy Projects on
Output Growth of Developing Countries – Testing for Conditional
Convergence

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Abstract

This empirical paper studies the impact of investment in renewable energy on total output using an augmented Solow model. The augmented Solow's model[1](Mankiw et al.) regards the textbook Solow's model [2](Solow) highly. Mankiw and others show including human capital and physical capital accumulation alters the theoretical modelling and empirical analysis of economic growth. The addition of human and physical capital accumulation increase the fit of Solow's model, explaining a majority of the cross-country variation. Growth in terms of GDP is a factor of many different parameters; omitted variable bias is unavoidable. This paper expands on the augmented Solow' model by Mankiw and others to include investment in renewable energy.

There is no consensus in the existing literature on impact of renewable energy and growth rates of output. The literature is very sensitive to the model misspecification. Chang(*see Chang et al. 2009*) has argued for the presence of a unidirectional causality between output growth and investment in energy[3]; Giraud on the other hand(*see Giraud G. & Kahraman Z. (2014)*), argues that energy consumption causes growth which validates the growth hypothesis. [4]. A more recent study conducted by Fan and Hao (*see Fan, W., Hao, Y. (2019)*) within China is in favour of renewable energy and gross domestic product per capita. A modest slowdown in growth rates will generate a significant boost in renewable energy consumption conditional on targeted foreign direct investment in the long run [5]. Fan and Hao look through the lens of growth rates and investment as triggers for renewable energy consumption. This paper rather has a focus on investment in renewable energy as a trigger for growth in developing countries.

This paper looks at whether the addition of a proxy for renewable energy investment lowers the coefficient on the initial level of output in the augmented solow's model. That is, does it accelerate the convergence speed. It uses a cross-country panel dataset of 90 countries. The dataset is a compilation of the Penn world Table, World Bank Data and International Renewable Energy Association(IRENA) datasets. The findings of this paper are that in the short run, the relationship between investment in renewable energy projects and output growth fails to hold; in the long run, an investment in renewable energy projects will benefit the developing countries more.

Key Findings

In the long run, there exists a statistically significant impact of investment in renewable energy projects on output growth rates. An investment in renewable energy would hence benefit developing countries more. This is conditional on the assumption that developed countries will attain steady and stagnant growth rates prior to developing nations. This relation passes all robustness checks conducted – Dataset check, Year fixed effects & A change in proxy for renewable investment. (Refer to section 2.5)

In the short run there is no relationship between the amount of investment in renewable energy projects and output growth. The growth rate of a country is not a statistically significant determiner of the amount of investment in renewable energy projects; in the short run. Countries with lower growth rates (Growth rates which fall below the regression line, refer to Figure 3.1) tend to have higher amount of investment in renewable energy projects.

Investment in renewable energy projects are lump sum development projects funded by developed countries or the state. The use of growth rate of investment in renewable energy projects is hence, fundamentally wrong. Hence, in order to access the impact of investment in renewable energy projects on developing countries an augmented solows model has been used. The coefficient of the augmented solows model can be manipulated to find the rate of convergence, i.e, in the selected sample of countries how fast are the countries converging to their steady states. (Refer to Solows 1956 for steady states) Adding controls for renewable energy investment, the coefficient reduces and the model becomes a better fit, implying that an investment in renewable energy, in the long run, will help countries reach their steady states faster. Developing countries will take a longer time to reach their steady states as compared to developed countries. An increase in the rate of convergence would speed up this process for developing countries due to increasing rate of returns. Based on the assumption that developed countries are further away from developed countries in terms of their output growth rates and would tend to experience a lower increasing rate of returns. An investment in renewable energy would hence benefit developing countries more.

Chapter 1

Background

The topic of economic growth has been of interest to economists. Structural change, standards of living, inequality are topics of development interest; developing countries are put at the forefront for this change. There exist differences between countries – differences in income levels, differences in opportunities, differences in growth rates; are few to name. I look at growth rates as an indicator for differences among countries in this study using an augmented solow's model regressing log growth rates on log of gross domestic product in 1970's with controls for human and physical capital accumulation and investment in renewable energy projects.

Solow [2], in 1956, initiated the study of economic growth. He used a standard neoclassical production function with decreasing returns to capital; rates of saving and population kept exogenous. Labour and capital were the determinants of the steady-state output per capita (level of output per effective unit of labour). Different countries reach different steady states – the population grows at different rates in different countries; The rate of saving is different for different countries.

Conditional convergence is the tendency of the poor countries to grow faster than the richer countries and converge to similar levels of income. There is however, a caveat, this converge is conditional on institutions being similar in the countries for which convergence is tested. In theory, two countries with different institutions will not converge. The caveat also highlights the role which the institutions play in economic development, as to grow at the rate of convergence, i.e catch-up, one must have similar institutions to the economy with better income levels.

This paper argues that investment in renewable energy projects will help the developing countries grow faster using an augmented solow's model. Investment in Renewable energy projects, is however, a must irrespective of the growth outcome to tackle the problem of Climate Change. The Earth has already warmed by 0.7°C since around 1900 and is committed to further warming over coming decades simply due to past emissions[8]. To understand the importance of a clean energy transition, it is important to understand the impact of climate change. Climate change will have adverse irreversible damages to everyone, in particular more serious impacts would be bore by developing countries.

While the most serious impacts of climate change will fall on the poorest countries, the developed world will be far from immune. Climate change will have some positive effects for a few developed countries for moderate amounts of warming, but is likely to be very damaging for the much higher temperature increases that threaten the world in the second half of this century and beyond if emissions continue to grow.
—Stern Review

1.1 Climate Change and Clean Energy Transitions

Drawing upon from the previous section, we can conclude that, climate change is an important topic of concern to people of all majors today. Economists, biologists, physicists, businessmen are looking at climate change from one lens or the other. A major contributor to climate change are green-house gasses; specially carbon-dioxide. The energy sector emits the highest levels of carbon dioxide as it burns coal in order to produce electricity. A clean energy transformation in the energy sector will help reduce the carbon emissions by a huge extend and help mitigate the impact of climate change. Developing countries are tapping into their abundant resources to generate clean electricity.

A clean energy transition will yield in reduced levels of carbon-dioxide and benefit the environment. This transition comes with an added cost to the governments of the developing countries. Through this paper, I wish to test if this cost(the investment to set up clean energy plants) which the government incurs would have a net positive effect on growth rates in developing countries. Will this form of investment help them grow faster? Will it slow down growth? The answer to this question is in the scope of this study.¹ Another popular argument in climate science literature is that investment in clean energy is widening the preexisting inequalities. This paper can be futher extended to see converge in HDI indicators to answer the same.

This paper has a focus on developing countries; in particular India. India is a country blessed with natural resources. The possibilities for clean energy generation is immense due to its geological advantages. It is important to first establish the current state of clean energy and India; looking at historic carbon emissions, literature on the current state of clean energy in India, and, a few climate trends; in order to gauge where India stands.

1.2 Renewable Energy Projects

The future of renewable energy is very promising. According to the IEA's new report on Clean Energy Transitions[9] in the greater Horn of Africa, A modern mix of clean energy can power an economy twice as big as today's with just 30% more energy while providing universal access and limiting emissions. A very ambitious but possible solution is to integrate the dual economies present in a nation into a national economy.

On a similar note, India is endowed with boundless potential in energy, particularly clean energy. India, at the conference of parties-26(COP), has promised to cut its emissions to net zero by 2070. The net-zero emission promise is set at the base of a strong policy formulation. What appears to be an aggregate concept at large, investment in renewable energy is perhaps the modern answer to rural and urban needs.

As policymakers, questions of inequality and development are necessities for the successful implementation of any policy. clean energy is a possible answer to the age-old question of power supply to rural India alongside being an obvious answer to the net emissions promise made by India. Thus, of concern to policymakers. The shift to clean energy is bound to happen – now or later. Will the effects of a shift to clean energy translate to the sustainable growth of development opportunities, devoid of inequalities or would it be a burden to the taxpayers of the nation?

¹To compute whether the cost incurred by governments, if renewable energy projects are funded by the state is more than the benefits of economic growth is not in he scope of this study.

Chapter 2

The Developing Country Perspective

What is often overlooked is the fact that a clean energy transition is capital intensive. A switch to clean production processes requires investment in technology and human resources. Developing countries are bestowed with an comparative advantage in labour and raw materials. Take India for example. India is a labour intensive country; its comparative advantage is its abundant labour supply. Despite having enormous potential in renewable energy sources, India is still dependant on foreign reserves of oil and coal to meet its energy demands. The maintenance, the gestation period and the initial investment required to set up a renewable energy project play against the comparative advantage of any developing country. Even if we drop the assumption of profit maximization it seems hard to convince a government as to why it would take up the tedious task of switch to clean energy for the sake of the environment. It would rather set up a car industry or an employment guarantee program to either promote industrialization or welfare. A strong case must be build for climate change and the effects of an renewable energy transition in order to pursue governments to invest in these projects. Developed countries play a major role in this transition; either in the form of threats or funds for projects.

The United Nations Environment Program has taken action to aid the developing countries with low-emissions and climate-resilient projects in the form of Green Climate Fund's (GCF). The Green Climate Fund's (GCF) is a unique approach taken by the United Nations Environment Program to limit or reduce greenhouse gas (GHG) emissions in developing countries, and to help vulnerable societies adapt to the unavoidable impacts of climate change.

The main aim of the GCF is to expand collective human action to respond to climate change. The Fund aims to mobilize funding at scale to invest in low-emission and climate-resilient development on our home planet. It allocates its resources to low-emission and climate-resilient projects and programmes in developing countries. The Fund pays particular attention to the needs of societies that are highly vulnerable to the effects of climate change, in particular Least Developed Countries (LDCs), Small Island Developing States (SIDS), and African States.[10] – UN Environment Programme

Developing countries will benefit largely from projects undertaken by the GCF as they are low-emission based and climate resilient. Projects initiated under GCF will also create awareness among the population regarding climate change.

2.1 Objectives

This study follows the landmark study “A Contribution to the Empirics of Economic Growth” by Mankiw, Romer and Weil from 1992. In this paper, I test for, empirically, if investment in clean energy transition is a statistically significant addition, and whether it helps developing nations grow faster, when compared with developed nations; Through this study I wish to understand if a switch from coal to clean energy is viable and will it yield in an economically beneficial transfer alongside improving the environment. This paper does not intend to devalue the impact of climate change and considers climate change a significant catastrophe for the future, for which a switch to clean energy is must and justified, rather tries to understand if in the future, by deviating from comparative advantage today, will the developing country receive a reward for the risk undertaken.

The primary objective of this paper is to determine whether an increase in investment in renewable energy projects will benefit the developing countries more than the developed countries. This paper will also attempt to calculate the speed of convergence before and after the addition of investment in clean energy.

2.2 Empirical Strategy

To make the connection between investment in renewable energy projects and growth rate in gross domestic product across developing and developed nations, an augmented solow’s convergence model with controls for human and physical capital accumulation was adopted. The solow model uses the following equation to study convergence.

The Solow model suggests the following regression to study convergence

$$\ln(GDP(t)) = (1 - e^{-\lambda t}) \ln(GDP^x) + e^{-\lambda t} \ln(GDP_{1970})$$

Here, λ stands for the convergence rate in time t . GDP^x is the steady-state production of output per effective worker which comprises of all the solow’s residual and human and physical capital accumulation.

Subtracting $\ln(GDP_{1970})$ from both sides

$$\ln(GDP(t)) - \ln(GDP_{1970}) = (1 - e^{-\lambda t}) \ln(GDP^x) - (1 - e^{-\lambda t}) \ln(GDP_{1970})$$

Substituting the value of GDP^x which is the steady-state production of output per effective worker which comprises of all the solow’s residual and human and physical capital accumulation, and adding controls for investment in renewable energy, we can formulate our econometric regression equation.¹

$$\begin{aligned} \ln\left(\frac{GDP}{Population}\right)_{2020} - \ln\left(\frac{GDP}{Population}\right)_{1970} = & \beta_0 + \beta_1 \ln\left(\frac{GDP}{Population}\right)_{1970} + \beta_2 \ln\left(\frac{I}{GDP}\right) \\ & + \beta_3 \ln(n + \gamma + \delta) + \beta_4 \ln(PrimarySchooling) + \\ & \beta_5 \ln(I_{RE}) + \beta_6 \ln(Employment_{RE}) + \epsilon_i \end{aligned}$$

$$\begin{aligned} GDP^x = & (1 - e^{-\lambda t}) \log(s_k) \\ & + (1 - e^{-\lambda t}) \log(s_h) \\ & + (1 - e^{-\lambda t}) \log(n + g + \delta) \end{aligned}$$

where s_k is share of physical capital accumulation and s_h is share of human capital accumulation. n stands for growth rate of working age population γ stands for growth rate of technology and δ stands for change in depreciation

2.3 Empirical Results

This section talks about the dataset and provides some descriptive statistics to motivate the analysis. The paper uses three cross-country panel datasets; The Penn world table provides with output, investment, depreciation and TFP growth rate data, The World bank dataset gives growth rate and population data for the robustness check and finally IRENA data for renewable investment and related proxies. provides us with information on invest. The educational attainment cross-country cross-sectional data is taken from the Barro-Lee which conducts it's own survey and is regarded highly for educational attainment data.

Table 4.1 Provides us with the percentage of people employed in the renewable energy sector. The rows corresponding to the red shade highlight developed countries; the rows shaded green highlight developing nations. With the exception of Brazil, all developed countries hire more people in the renewable energy sector. This paper uses employment as a proxy for investment in renewable energy as more the percentage of population employed in the renewable energy sector the stronger the signal of higher investment n renewable energy. This showcases how the developed countries are more invested in the idea of renewable energy transition and mitigating climate change effects.

In the short run there is no relationship between the amount of investment in renewable energy projects and output growth. The growth rate of a country is not a statistically significant determiner of the amount of investment in renewable energy projects; in the short run. Countries with lower growth rates (Growth rates which fall below the regression line, refer to Figure 3.1, Table 4.1) tend to have higher amount of investment in renewable energy projects. From Table 4.1

it can be seen that there is no effect of investment in renewable energy projects on GDP growth rates. There exists a statistically significant impact of invest in renewable energy projects on output growth at the one percent level of significance, however, subsetting for High and Low growth rates, the relationship does not hold. Figure 4.3 also shows that there is no relation among growth rates and investment. The use of growth rate of investment in renewable energy projects is also a fundamentally wrong indicator as these projects are usually funded in intervals and with one interval having very high funding, say for a large project as compared to others. Based on the following evidence, I reject the idea of any relationship between investment in renewable energy projects and GDP growth, at least in the short run.

In the long run, countries converge to their respective steady states(Solow 1964). An addition of two variables; investment in renewable energy and share of employment in the renewable energy sector; further lowers the coefficient on the initial level of income. That is, it accelerates the convergence speed. From the Penn World Tables 10.0, the addition of investment in renewable energy project lowers the coefficients by 0.01, increasing the fit of the model by 0.017. The coefficient on the initial level of income is now statistically significant as well at the 10% level of significance. From the World bank Data, the addition of investment in renewable energy project lowers the coefficients by 0.007, increasing the fit of the model by 0.012. The coefficient on the initial level of income is now statistically significant at the 5% level of significance.

Based on the above evidence, we can argue that an increase in investment in renewable energy projects will yield in higher growth rates of GDP for developing countries as compared to developed countries. The foundation of this argument is set on the fact that developing countries will take a longer time to reach their steady states as compared to developed countries. An increase in the rate of convergence would speed up this process for developing countries due to increasing rate of returns. Based on the assumption that developed countries are further away from developed countries in terms of their output growth rates and would tend to experience a

lower increasing rate of returns. An investment in renewable energy would hence benefit developing countries more.

2.4 Conclusion

The relationship between investment in renewable energy projects is sensitive to model misspecifications. This paper proves that there exists a statistically significant impact of investment in renewable energy projects on output growth rates. An investment in renewable energy would hence benefit developing countries more. This is conditional on the assumption that developed countries will attain steady and stagnant growth rates prior to developing nations. This relationship holds true only in the long-run as achieving respective steady states is a long term concept.

2.5 Robustness Checks

This paper conducts a three robustness checks. To begin with, I have changed the dataset used for growth rates and initial GDP values in order to eliminate the question of a bias arising from growth rates and GDP dataset selection. Growth rate from Penn world Table and World bank show a difference in coefficients but the story is the same – Adding proxies for renewable energy investment further lowers the coefficient on the initial level of income. That is, it accelerates the convergence speed. In addition, I have run the regression analysis with year fixed effects and average growth across years. Both provide similar results. The same goes with the final robustness check of adding an additional variable, in this case, employment alongside investment in renewable energy projects as a proxy of investment in renewable energy projects; the value of R^2 reduces, this however, does not change the interpretation of the coefficient on the initial level of income. This regression model has passed the above robustness checks.

Chapter 3

Figures

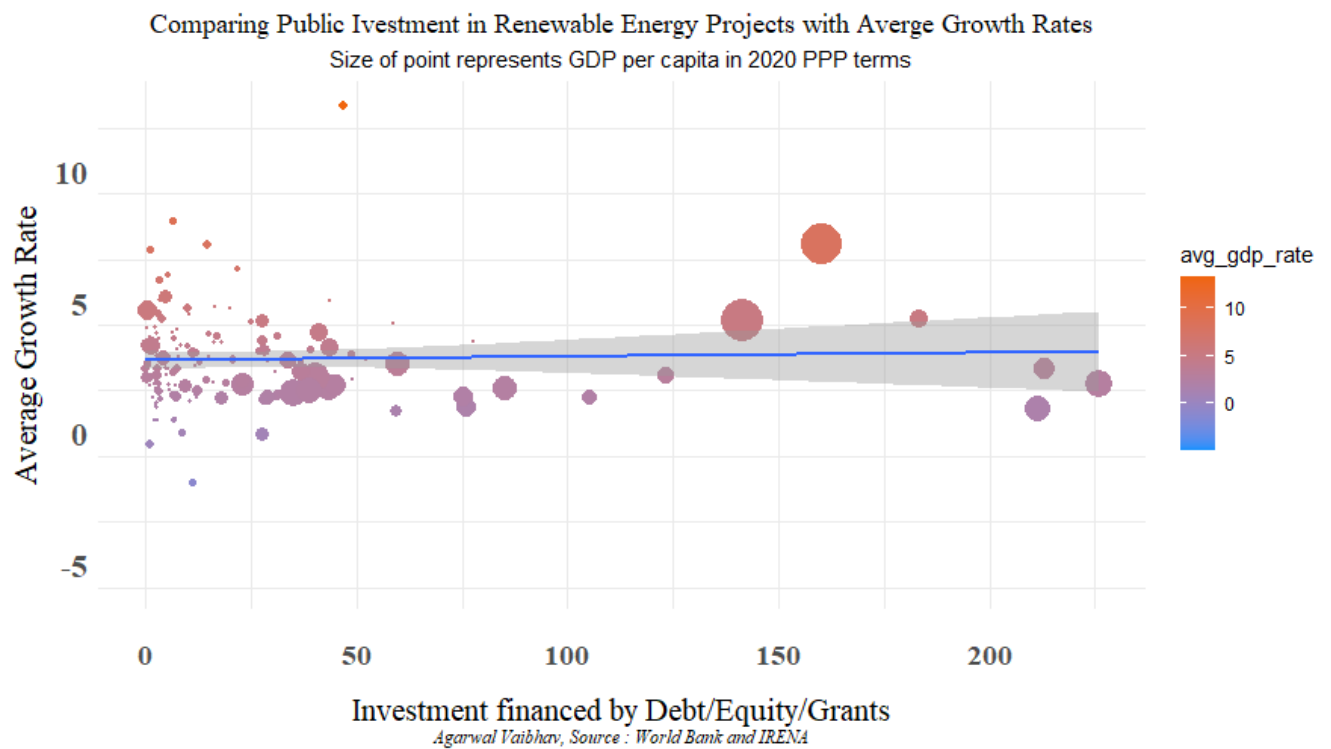
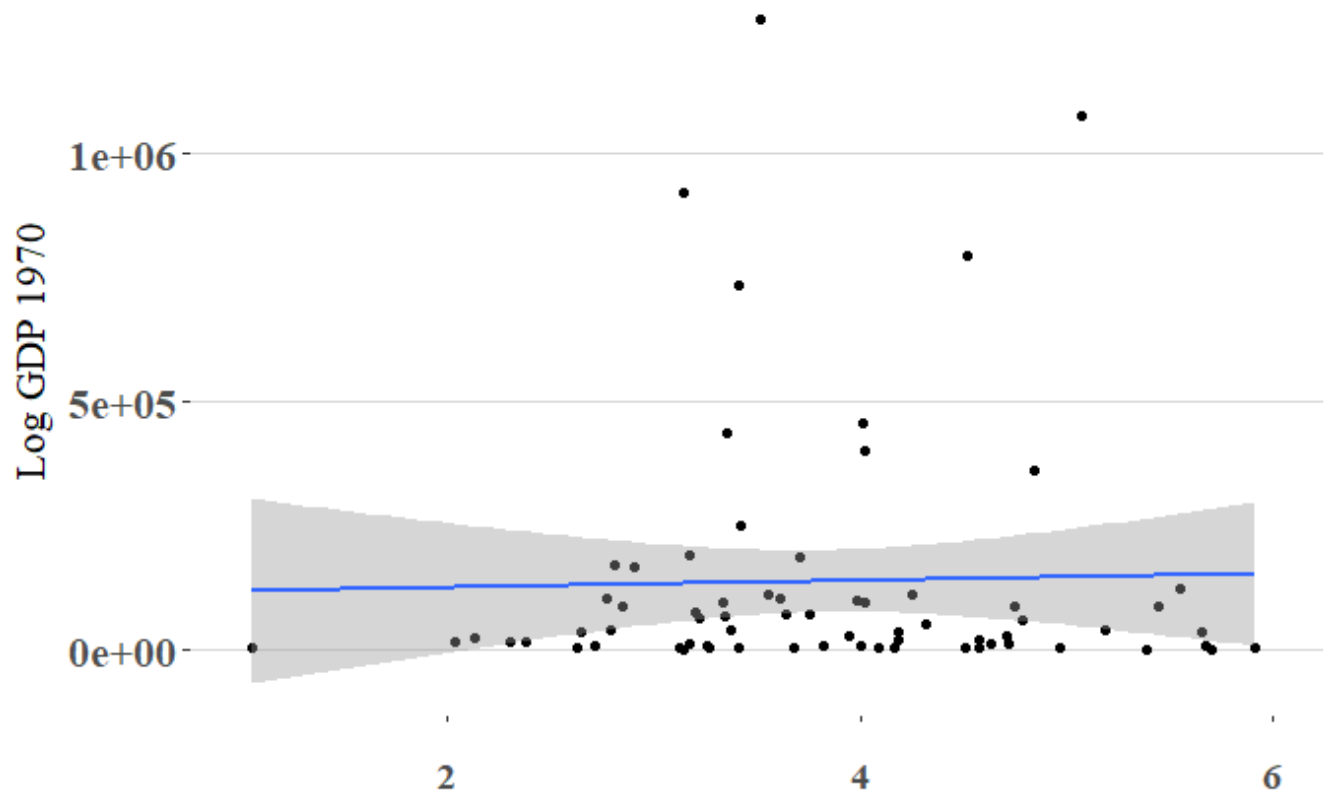
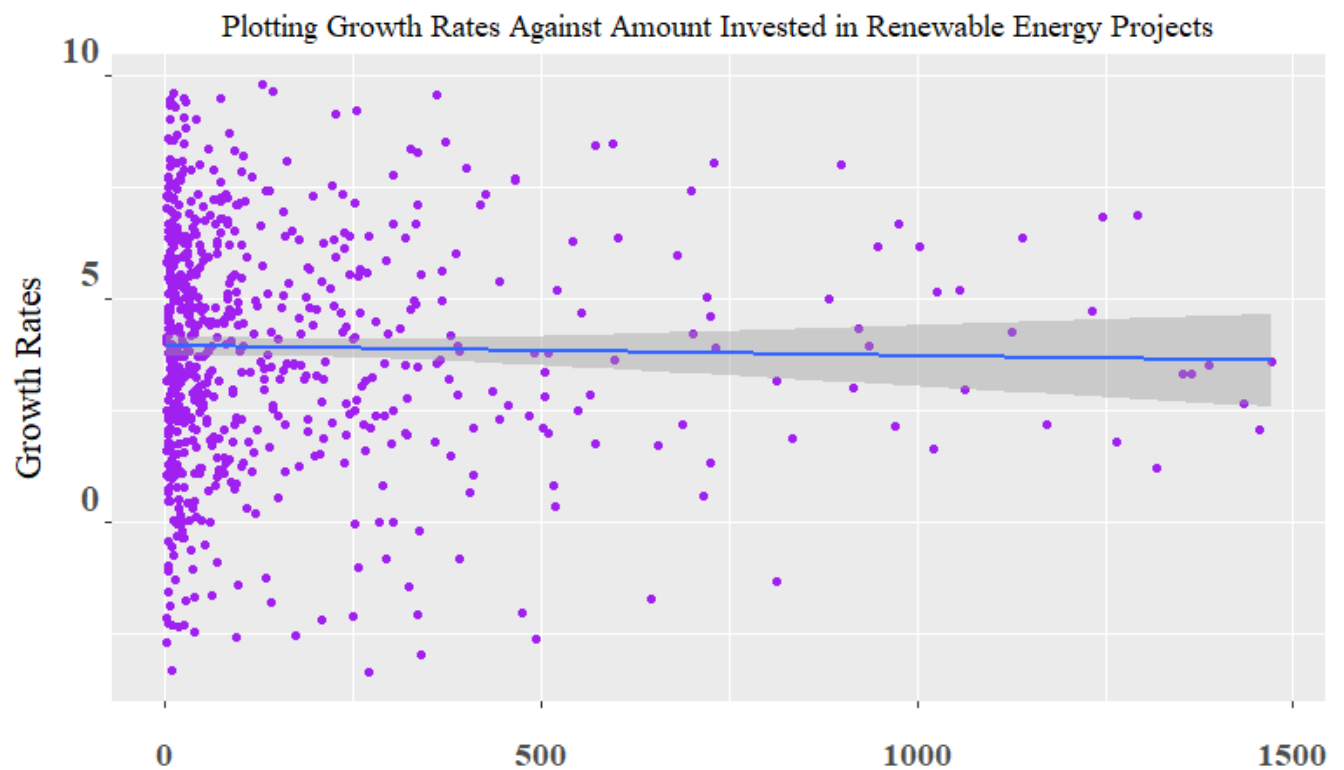


Figure 3.1: Capturing Developed and Developing Countries Growth Rate With Respect to Investment in clean Energy

Test for Unconditional Convergence



Average Growth Rate
 Source : NCRB Dataset- Crime in India Report



Amount Invested in Renewable Energy Projects
 Agarwal Vaibhav, Source : World Bank and IRENA

Chapter 4

Tables

Table 4.1: Impact of Investment in Renewable Energy Projects on GDP growth rates

	<i>Dependent variable:</i>				
	All	High G	Low G	High RI	Low RI
	(1)	(2)	(3)	(4)	(5)
Investment	0.007*** (0.003)	0.003 (0.002)	0.001 (0.003)	0.035* (0.021)	0.009* (0.005)
Constant	3.382*** (0.216)	5.621*** (0.178)	1.345*** (0.193)	0.743 (1.980)	3.353*** (0.249)
Observations	893	503	390	484	409
R ²	0.008	0.003	0.001	0.006	0.008
Adjusted R ²	0.007	0.001	−0.002	0.004	0.006
Std. Error	2.700	1.568	1.703	2.686	2.718
F Statistic	6.960***	1.295	0.319	2.789*	3.269*

Note:

*p<0.1; **p<0.05; ***p<0.01

Agarwal Vaibhav

Table 4.2: Percentage of People Employed in the clean Energy Sector

Red – Developed Countries; Green – Developing countries

Agarwal Vaibhav , Source: International clean Energy Association (IRENA) and World Bank

Rank	Country	Employment Rate in cleans
3	Denmark	1.20
5	Sweden	0.86
6	Brazil	0.85
8	Finland	0.71
9	Germany	0.66
14	China	0.54
19	Norway	0.49
20	Netherlands	0.47
26	Spain	0.35
31	Indonesia	0.31
32	Thailand	0.31
34	Canada	0.29
39	United Kingdom	0.23
40	Japan	0.22
62	India	0.09
63	Ghana	0.08
68	Pakistan	0.07
70	Kenya	0.06
91	Sri Lanka	0.01
95	Botswana	0.006
97	Nepal	0.0051

	Dependent variable:	
	log(Growth rate of Expenditure in GDP in real terms(Average GDP growth rate), Source - Penn World Table 10.0')	
	(1)	(2)
log(The GDP at constant dollars in 1970(Year of reference post oil crises of 1970), Source - Penn World Table 10.0')	-0.036 [*] (0.019)	-0.026 (0.017)
log(n+gamma-delta(gamma assumed 0.02 where value is NaN')	0.179 ^{***} (0.040)	0.176 ^{***} (0.040)
log(Average years of secondary schooling - Proxy for Human Capital Accumulation (Source: Barro, Robert and Jong-Wha Lee, 2013')	-0.020 (0.077)	0.020 (0.068)
log(Average Public Investment In Renewable Energy Projects In US Dollars(Millions), Source - IRENA')	0.029 (0.030)	
log(Percentage of Persons Employed in Renewable Energy Sector by working age population of country, Source- IRENA(Employed) and World Bank(WAP')	0.008 (0.024)	
log(The average share of Investment in GDP, Source - Penn World Table 10.0')	0.109 (0.104)	0.095 (0.099)
Constant	1.838 ^{***} (0.387)	1.677 ^{***} (0.290)
Observations	66	66
R ²	0.283	0.266
Adjusted R ²	0.210	0.218
Residual Std. Error	0.225 (df = 59)	0.224 (df = 61)
F Statistic	3.884 ^{***} (df = 6; 59)	5.524 ^{***} (df = 4; 61)
Note:	* p<0.1; ** p<0.05; *** p<0.01	

Dependent variable:		
	log(Average growth rate of GDP PPP Adjusted, Source - World Bank Data)	
	(1)	(2)
log('The GDP at constant dollars in 1970(Year of reference post oil crises of 1970), Source - Penn World Table 10.0')	-0.040** (0.020)	-0.047** (0.023)
log('n+gamma+delta(gamma assumed 0.02 where value is NaN)')	0.173*** (0.048)	0.169*** (0.049)
log('Average years of secondary schooling - Proxy for Human Capital Accumulation (Source: Barro, Robert and Jong-Wha Lee, 2013)')	-0.200** (0.082)	-0.215** (0.092)
log('Average Public Investment In Renewable Energy Projects In US Dollars(Millions), Source - IRENA')		-0.016 (0.036)
log('Percentage of Persons Employed in Renewable Energy Sector by working age population of country, Source- IRENA(Employed) and World Bank(WAP)')		0.030
log('The average share of Investment in GDP, Source - Penn World Table 10.0')	0.112 (0.119)	(0.029)
Constant	2.171*** (0.348)	2.475*** (0.465)
Observations	66	66
R ²	0.335	0.347
Adjusted R ²	0.291	0.281
Residual Std. Error	0.268 (df= 61)	0.270 (df= 59)
F Statistic	7.681*** (df= 4; 61)	5.233*** (df= 6; 59)
Note:	* p<0.1; ** p<0.05; *** p<0.01	

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